Rev	ECO #	Date	Revision Description
J	0034158574	Mar 14, 2022	• 6.1. Qualification Unit: clarified vendor's responsibilities • 7.1. Procedure for equipments and methods: Clarified need to highlight in case of schedule risks. • 7.2. Workflow for equipments and methods: Added PRQ steps • 7.3. Workflow for material: split scenarios of new material qualification and material change qualification • 8.3. Evaluation and Qualification for process: clarified thermal stability conditions for FPC • 8.3.9 Cleaning after cavity cut/outline cut/singulation: added guidelines • 8.4. Solder mask: added flux resistance and high speed ball shear for CQRA testing • 8.6.1 AFVI, VRS, artificial intelligent: added items related to AI qualification • 9.1.5. new EMI shield material: added

PCB FPC LINE QUALIFICATION GUIDELINES

080-02647

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1. Purpose

The purpose of this specification is to establish acceptability requirements for any new and existing equipments (including tooling/jig/fixture), new manufacturing methods (process parameter, materials and chemical, process flow) and new material not used in any Apple's rigid, rigid-flex and flex printed circuit boards before. Those equipments, methods and materials shall be qualified by the procedures defined in these guidelines.

2. Scope

This specification applies to the quality requirements of rigid, rigid-flex and flex printed circuit boards.

3. Responsibilities

The rigid, rigid-flex and flex PCB fabrication supplier will be responsible for the fabrication of PCB's and the qualification of new and existing equipments in compliance with this specification unless an authorized Apple representative has provided deviations from these requirements in writing.

4. References

#	APN	Title
1	062-0073	Rigid Printed Circuit Board Specification
2	099-01674	Rigid and Rigid Flex Printed Circuit Board Quality Control Requirements
3	080-2265	Flexible Printed Circuit Board Specification
4	080-01409	Apple Bare FPC Critical Process Parameters
5	080-01072	Apple Bare FPC Quality Control Plan Requirement
6	080-02999	Apple Bare FPC General Continuous Quality Reliability Assurance Requirement
7	080-02882	NPI OK2Ship Report for Bare FPC

5. Definition

POR equipment: Plan of Record machine that finished qualification
POR material: Plan of Record material that finished qualification
PCB: Printed Circuit Board (including rigid, rigid-flex, flex and substrate)

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6. General

6.1. Qualification Unit

Vendor must qualify individually each equipment, manufacturing method and material.

- PCB/FPC vendor should use **one type of machine**[1] **per process per project APN per layer**. If not feasible, vendor must consult Apple Fabrication team, and receive written approval from Fabrication team DRI before use in an actual Apple build or MP. Qualification steps include at the minimum UPD 1, 2, CQRA and 3x3 LQ lots. Refer to scenario 3 in qualification scenario table.
- PCB/FPC vendor is responsible to control materials and raw materials throughout project NPI until its end of life, and must ensure no process changes at material vendor sites. Any changes at material level is considered to create a "new material". Vendor must follow new material workflow for qualification.

6.2. Risk Assessment and qualification scenarios

A. Equipments and methods: Refer to following pages for details
B. Material qualification: vendor must publish proposal for Evaluation contents (methods, quantity, criteria). Apple DRI must review and approve before proceeding.

6.3. Qualification Procedure

Refer to following pages for details.
Every steps must be done before MP.

6.4. Test Specimen

- Test specimen for each test item are detailed "Product Quality and Capability Confirmation" and "CQRA" sections for each process.
- When test coupons are required, test coupon design and materials must be the same as actual product and have same layer count in stack-up for via integrity evaluation. In any other cases, vendor must use actual product. Design of the actual product to use is determined by design compatibility. Refer to procedure pages for compatible design definition.

6.5. Equipments, Chemicals List

From P1 and at every build before input, vendor must publish a list of qualified equipments, chemicals and material to Apple DRI, including DFM, SQE and GSM.
Vendor must update the list at every build, reflecting all changes, additions, and removals from previous builds.
Vendor must publish the final list before MP.

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[1] A type of machine is a combination of all physical characteristics including but not limited to equipment's vendor, model, design, internal components's types, dimensions, mechanical construction, etc. (e.g. bath dimensions, chemical used, nozzles locations or directions, etc.) as well as software, and electrical settings. If any characteristic is different, then it's considered as different type.

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7. Qualification Procedure

7.1.Scenarios for equipments or manufacturing methods

Below table specifies qualification steps requirements based on risk assessment, for **each process and each project APN**. For each scenario, requirements are different because risk level might be different. Requirements apply to any changes after P2, except scenario #9. PCB/FPC vendor must finish all steps and receive approval from Apple SQE **before next build's input** (E.g. EVT, etc.). If there are any concerns or conflicts between LQ schedule and NPI build schedule, PCB/FPC vendor must highlight to Apple SQE as soon as possible to discuss about risk mitigation and LQ plan.

#	Location	When to use this equipment or method?	Criteria for risk assessment					Qualification steps		
			Equipment previously used at the exact same location within the same factory	UPD *1 already done on a compatible design *3 related to that process	Equipment used in P1/ P2 and EVT?	Equipment exactly same as the ones already qualified? *4	Process recipe / parameters / method same as the ones already qualified? *5	UPD *1 step 1	UPD *1 step 2 *2	Line qualification lots *6
1	Same PCB/ FPC vendor & location (room or floor) as POR	EVT or anytime after EVT	New buy or equipment relocated within same room or floor	UPD done	No	Same	Same as POR	Must	Must	Must 3 lots = 1 lot per day * 3 days
2			Ignore this if design used for UPD is not compatible	Design not compatible	Ignore this if design used for UPD is not compatible	Same	Ignore this if design used for UPD is not compatible	Must	Must	Must 3 lots = 1 lot per day * 3 days
3			Ignore this if equipment is different	Ignore this if equipment is different	Ignore this if equipment is different	Different	Ignore this if equipment is different	Must consult DFM. DFM do not recommend using different equipments on one project. Minimum requirement is 9 lots = 3 lots per day x 3 days. DFM will decide additional requirements based on discussion with vendor		
4			Yes	UPD done	Yes	Same	Different *5	Must	Must	Must 9 lots = 3 lots per day * 3 days
5	Different from POR		Floor, building, factory or campus different	Ignore this if factory is different	Ignore this if factory is different	Ignore this if factory is different	Ignore this if factory is different	Must	Must	Must 9 lots = 3 lots per day * 3 days
6	Same PCB/ FPC vendor & location as POR	EVT	Yes	UPD done	Yes	Same	Same as POR	No need	No need	No need
7		EVT or anytime after EVT	Yes	UPD done	No	Same	Same as POR	Must	No need	No need
8		Anytime after EVT	Yes	UPD done	Used in P1 or P2, but not EVT	Same	Same as POR	Must	No need	No need
9		P2	Yes	Ignore this if panel size or layout is different	Ignore this if panel size or layout is different	Same	Different panel size or layout	Must	Must	No need

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Definitions and notes:

***1 Unit Process Development (UPD)**

- Before qualification, each individual line or machine must have UPD, once per machine and per compatible design.
- Every time a machine is relocated, it needs to have UPD again even if the relocation is within the same factory (e.g. floor 1 to floor 4)
- UPD has 2 steps:
  - **Step 1**: build small quantity of DoE parts to set nominal process parameters and compare the product quality. It should not be statistically significantly worse than qualified machines.
  - **Step 2**: Build again small quantity of DoE parts at process parameter corners based on nominal settings from step 1. Quality must be within specifications.
- Items to confirm at each step include items detailed in "Product Quality and Capability Confirmation" table on each process section, not CQRA.
- UPD parts must go through POR intent process flow. Vendor shall not skip process steps.
- Machines must be fully loaded as MP, when vendor does UPD. Actual products or dummy panels can be used to replicate full loading. Including all chemical or wet process that use immersion (no need for the ones that use only spray), and following mechanical or dry process: Plasma dry desmear, Hot press laminator, Cover layer attach+pressing, Stiffener/EMI shielding attach+baking, Soldermask Pre-curing, Soldermask Post thermal curing, UV curing, Soldermask Plasma cleaning.

***2 Exceptions for UPD step 2**

The following processes need only UPD step 1: AOI, AVI, electrical test, inkjet legend printing, outline cut/singulation using punch, coverlay cut using punch, and panel slitting using punch.

***3 Compatible design**: A design with any of the following features: 1) layer count equal or higher, 2) line/space equal or smaller, 3) pad size equal or smaller, 4) laser/NC drill via size equal or smaller, 5) aspect ratio equal or higher, 6) geometry tolerance equal or tighter.

Not compatible design: A design with any of the following features.
Equipments impacted by these features need to run through UPD.

- different dielectric or copper materials or with different properties (e.g. thickness, PI color, resin content, etc.) or manufacturing sites (including raw materials), or different model name, including suffix and line up number (e.g.ABC-1234_2 and ABC-1234-T, etc.)
- different solder mask or coverlay materials,
- different surface finish.

***4 Equipment exactly the same as qualified equipment**

This includes all physical characteristics including but not limited to equipment's vendor, model, design, internal components's types, dimensions, mechanical construction, etc. (e.g. bath dimensions, chemical used, nozzles locations or directions, etc.) as well as software, and electrical settings. If any characteristic is different, then it's considered as different type.

***5 Scope of any changes in recipe, process parameters, or manufacturing methods**: includes but not limited to: panel layout or size, glass mask.

Any changes at material level is considered to be "new material". Vendor much follow new material workflow for qualification. This includes but is not limited to: changes of material vendor, material vendor process/line/equipment/factory, raw material vendor process/line/equipment/factory

For bonding sheet, coverlay punching die: any change from soft tool to hard tool after P2, and any change of cavity quantity per tooling after EVT.

Exceptions: below features are not taken into account for risks assessment:

- Copper pattern, via count, via location and outline design change in Apple ODB++.
- Electrical testing jig / fixture design change in Apple ODB++.
- DES etching or flash etching line speed adjustment to meet trace width nominal.
- ENIG nickel and gold parameter adjustment to meet thickness nominal.
- Inspection machines with MSA (Measurement System Analysis) confirmed and showing no difference with POR machines statistically.
- Tightened process parameters window.

***6 Line qualification lots**

- Vendor builds actual products. LQ lot size must be the exactly the same as MP lot size.
- Items to confirm include items detailed in "Product Quality and Capability Confirmation" and "CQRA" tables on each process section.
- After CQRA is finished and if quality of these lots is confirmed acceptable and process is qualified for the build as a result, then these lots can be used and shipped as POR parts, with NPI Ok2ship and full CQRA validated and approved.

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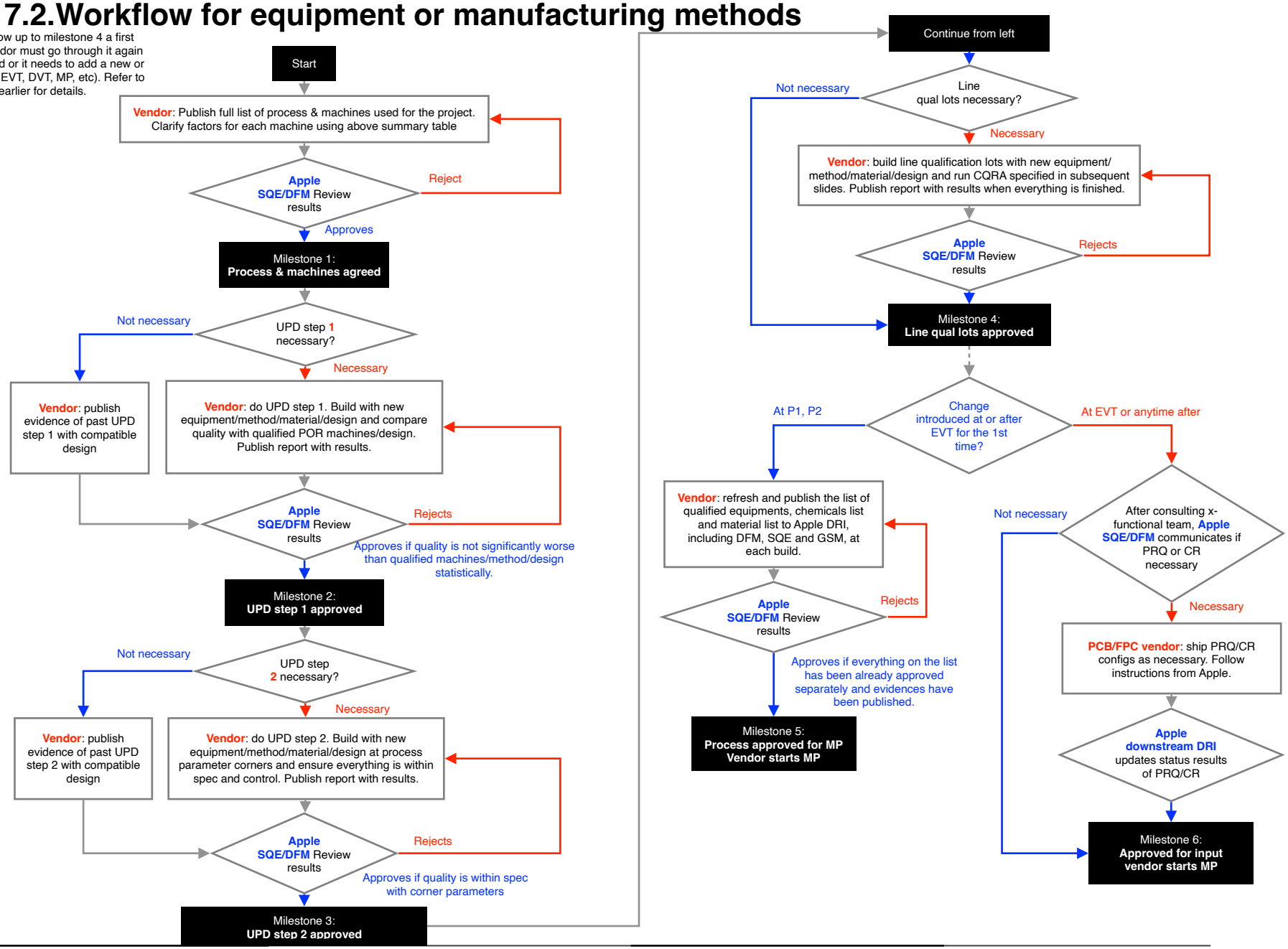
7.2.Workflow for equipment or manufacturing methods

Vendor goes through this flow up to milestone 4 a first time before input at EVT. Vendor must go through it again every time process is changed or it needs to add a new or existing equipment (e.g. after EVT, DVT, MP, etc). Refer to scenario table on earlier for details.



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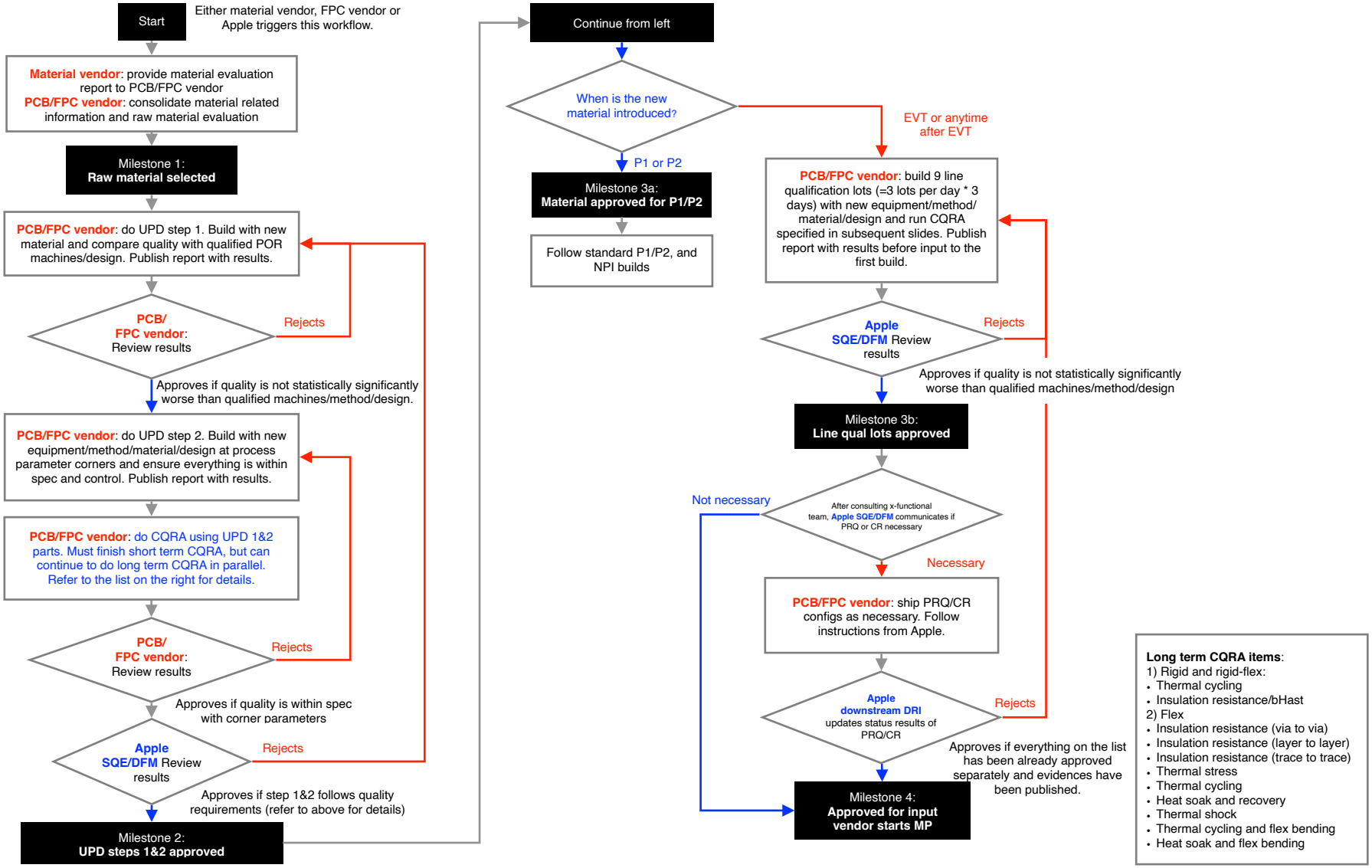
7.3.Workflow for materials

7.3.1.Workflow for new materials

New material refers to any materials not used in vendor's factory, including but not limited to changes to POR materials.



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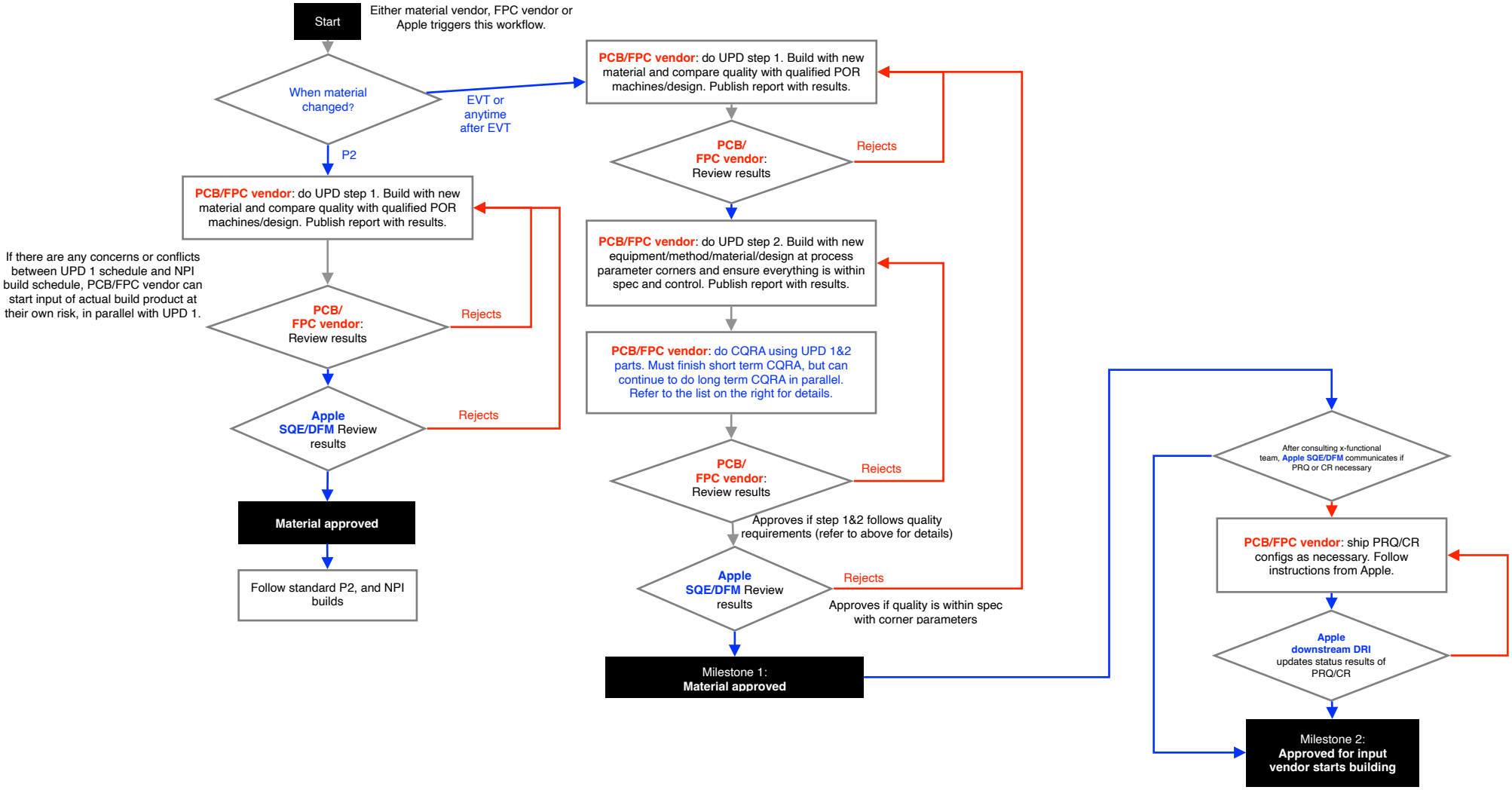
7.3.2.Workflow for material change

"Material change" refers to the change of materials on a specific APN at or after P2, but already qualified on other APN built in the same factory. Before P2, PCB/FPC vendor can change the material without following below workflow.



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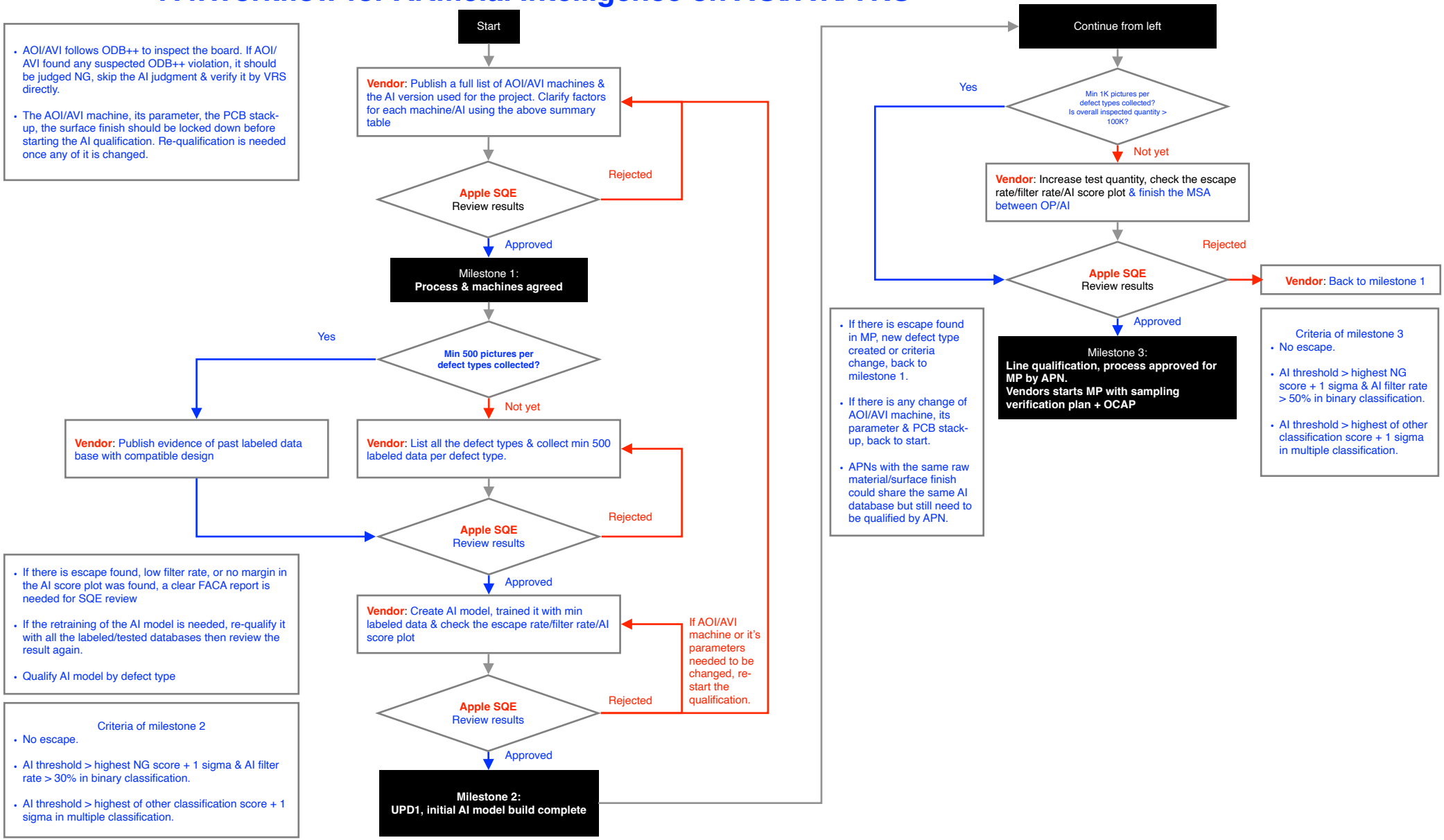
7.4.Workflow for Artificial Intelligence on AOI/AVI/VRS



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7.5.Equipments and Chemicals List Template

Vendor Name	
Factory/Site	
Project code	
APN	

Layer	**Process**		**Equipment**				**Chemicals (if applicable)**		**Remark**
	Main process	**Sub process**	**Factory code**	**Equipment code**	**Vendor name**	**Model code**	**Vendor name**	**Model code**	

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7.6.Material List Template

Vendor Name	
Factory/Site	
Project code	
APN	

Layer	**Material**		**Remark**
	Vendor name	**Model code**	

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8. Evaluation and Qualification Items for process

8.1.Micro Via Formation

8.1.1.Brown oxide for Direct laser

8.1.1.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Surface discoloration	Working Panel	Check 3 working panels/lot	No scratch, discoloration, water stain
Uniformity of color level (LAB value)	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Each value (max-min)/(2*avg)<10%
Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Overhang<10um
Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%
Wedge	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Wedge<10um

8.1.1.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.1.1.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.1.2.Black oxide for Direct laser

8.1.2.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Surface discoloration	Working Panel	Check 3 working panels/lot	No scratch, discoloration, water stain
Uniformity of color level (LAB value)	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Each value (max-min)/(2*avg)<10%
Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Overhang<10um
Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%
Wedge	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Wedge<10um

8.1.2.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.1.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.1.3.Laser drilling

8.1.3.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Laser via position accuracy	Working Panel	Check all the via on 3 working panels/lot by AOM or hole AOI	Spec: <minimum annular ring size, Cpk>1.33
Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Rigid, rigid-flex: Overhang<10um Flex: follow QCP and 080-2265
Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%
SEM/EDX to check the bottom of via from top view	Working Panel	Top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No smear residue

8.1.3.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (via to via)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.1.3.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.1.4.Plasma dry desmear

8.1.4.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount (weight loss) uniformity	Test Coupon	Check top/bottom side and corner/center position for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
SEM/FIB after Cu plating to check the bottom of via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No voids, smear residue
SEM/EDX after desmear to check the bottom of via from top view	Working Panel	Top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No smear residue
Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Rigid, rigid-flex: Overhang<10um Flex: follow QCP and 080-2265
Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%
Wedge (bare FPC: etch back at via bottom)	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Wedge<10um

8.1.4.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (via to via)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.1.4.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.1.5.Liquid desmear

8.1.5.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount (weight loss) uniformity	Test Coupon	Check top/bottom side and corner/center position for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
SEM/FIB after Cu plating to check the bottom of via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No voids, smear residue
SEM/EDX after desmear to check the bottom of via from top view	Working Panel	Top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No smear residue
Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Rigid, rigid-flex: Overhang<10um Flex: follow QCP and 080-2265
Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%
Wedge (bare FPC: etch back at via bottom)	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Wedge<10um

8.1.5.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (via to via)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.1.5.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.1.6.Overhang Cu flash etching/Cu foil etching

8.1.6.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Etching depth in the bottom of via	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Etching depth (max-min)/(2*avg)<10% and within SPC control limit
Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Rigid, rigid-flex: Overhang<10um Flex: follow QCP and 080-2265
Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%

8.1.6.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.1.6.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.1.7.UV cleaning for FPC

8.1.7.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Adhesion	Test coupon	Equipment: Shimadzu AGX, Location: 9points/PNL (Left, Center, Right 3x3=9)	>0.1N/cm2

8.1.7.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Hot bar
- Reflow

8.1.7.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.Pattern formation (Subtractive, mSAP and FPC Button Plating)

8.2.1.Electroless Cu plating

8.2.1.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
FIB to check E'less Cu thickness on via wall and bottom	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	Follow 062-0073, thickness > 0.2 μm
E'less Cu plating thickness on surface (Weight gain)	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Plating thickness or weight gain (max-min)/(2*avg)<10% and within SPC control limit

8.2.1.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (via to via)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.1.3.QMS

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8.2.2.Shadow/Black Hole

8.2.2.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
SEM/FIB after shadow line (after inline soft-etching) to check the bottom of via	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No voids, carbon residue
Resistance checked by via chain coupon	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Resistance (max-min)/(2*avg)<10% and within SPC control limit

8.2.2.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (via to via)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.3.Flash Cu plating

8.2.3.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Cu plating thickness uniformity, surface Cu	Dummy Panel	Check top/bottom side and 25 locations/side for uniformity by lot	Cu thickness (max-min)/(2*avg)<10%
Cu plating thickness, surface and via wall	Working Panel	Cross section top/bottom side, corner/center position and highest/ lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Within SPC control limit
Resistance checked by via chain coupon	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Resistance (max-min)/(2*avg)<10%

8.2.3.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (via to via)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.3.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.4.Pre-treatment for Dry film lamination

8.2.4.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit
Roughness uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Roughness (max-min)/(2*avg)<10%

8.2.4.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.4.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.5.Dry film exposure, Direct imager for “Cu trace”

8.2.5.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Trace width, pad size	Working Panel	Check top/bottom side and operation/center/pump side by lot	Cpk>1.33
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit
Alignment accuracy	Test Panel	Use test panel to check alignment accuracy by lot	Spec: <minimum annular ring size, Cpk>1.33

8.2.5.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Insulation resistance (trace to trace)

8.2.5.3.QMS

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8.2.6.Dry film developing

8.2.6.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Trace width, pad size	Working Panel	Check top/bottom side and operation/center/pump side by lot	Cpk>1.33
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit

8.2.6.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.6.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.7.Electrolytic Cu pattern/via-fill plating

8.2.7.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Cu plating thickness uniformity, surface Cu, side wall, bottom thickness, dimple depth	Working Panel	Check top/bottom side and 25 locations/side for uniformity by lot	Cu thickness (max-min)/(2*avg)<10%
Depth of dimple and height of protrusion on via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Cpk>1.33, dimple or protrusion <10um
Via filled void	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	No void allowed
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit
Plated Cu elongation	Plated Cu foil	Check elongation by lot	Elongation >20%
SEM/FIB after Cu plating to check the bottom of via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No voids, residue

8.2.7.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Insulation resistance (via to via)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.7.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.8.Via paste filling

8.2.8.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Void ratio	Test Coupon	Check 4 via at 5 locations (total 20 via) within 1 single panel (4 corners and center) per lot	≤ 20%
Filling ratio	Test Coupon	Check 4 via at 5 locations (total 20 via) within 1 single panel (4 corners and center) per lot	90~160%
IMC length ratio	Test Coupon	Check 4 via at 5 locations (total 20 via) within 1 single panel (4 corners and center) per lot	≥ 80%
Paste flow out & overflow	Working Panel	Check 100% of via within 1 single panel per lot	No paste flow out

8.2.8.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Heat soak
- Thermal Shock
- Thermal cycle
- Reflow
- Hot Oil
- Insulation resistance (trace to trace)
- Insulation resistance (via to via)
- Dielectric withstanding

8.2.8.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.9.Dry film stripping

8.2.9.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit

8.2.9.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.9.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.10.Annealing for Cu

8.2.10.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Trace width, pad size	Working Panel	Check top/bottom side and operation/center/pump side by lot	Cpk>1.33
Depth of dent on Cu surface after flash etching	Working Panel	100% AOI to check dent for all the lots	No dent because of flash etching
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit

8.2.10.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.10.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.11.Seed Cu flash etching

8.2.11.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Trace width, pad size	Working Panel	Check top/bottom side and operation/center/pump side by lot	Cpk>1.33
Depth of dent on Cu surface after flash etching	Working Panel	100% AOI to check dent for all the lots	No dent because of flash etching
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit

8.2.11.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.11.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.12.Subtractive Development-Etching-Stripping

8.2.12.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Trace width, pad size	Working Panel	Check top/bottom side and operation/center/pump side by lot	Cpk>1.33
AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit

8.2.12.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.12.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.13.AOI, VRS, artificial intelligence

8.2.13.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Gage R&R	Working panel	10 Panels/lot All different dielectric material should be tested.	Kappa > 0.9
Defect Escaping	Working panel	10 Panels/lot All different dielectric material should be tested.	0%
AI filter rate	Working panel	Review overall filter rate	Filter rate > 50%
AI escape rate	Working panel	Compare defect quantity captured by 1) VRS after AOI using AI and 2) VRS after AOI not using AI (including ODB++ mismatch)	• No escape from AI. Check each defect type. • Labeled picture > 1K per defect type. Pass pictures : total defect picture ~ 50 : 50. Overall inspected quantity > 100K.
AI score plot	Working panel	Review AI score distribution of binary classification & multiple classification	• Min 1K pictures per defect type in trained database & total 100k pictures inspected for qualification are needed. • AI threshold > highest NG score + 1 sigma in binary classification. • AI threshold > highest of other classification score + 1 sigma in multiple classification.
Kappa	Shipping Strip	Kappa, κ= (Po-Pe)/(1-Pe) 5 pcs serious defect + 5pcs slight defect per defect type + 20pcs good pcs.	κ > 0.9 No defect escape

Definitions:

- Filter rate = quantity judged by AI correctly / suspected NG quantity judged by AOI or AVI
- Escape rate = defect quantity escaped from AI / overall AI inspection quantity
- Po is the relative observed agreement among raters, and Pe is the hypothetical probability of chance agreement

8.2.13.2.CQRA

Not required.

8.2.13.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.2.14.AOR, AOS (for short repair only)

Note: Specific approval from Apple DFM/SQE will be needed for Repairing position, quantity, size

8.2.14.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Trace width after repairing	Working Panel	10% sampling and checking AOR repaired trace width	Cpk>1.33
DCR	Working Panel	10% sampling and checking AOR repaired trace width	No significant difference statistically

8.2.14.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Thermal cycle
- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.2.14.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.Dielectric lamination formation

8.3.1.Brown oxide for dielectric lamination

8.3.1.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Surface discoloration	Working Panel	Check 3 working panels/lot	No scratch, discoloration, water stain
Uniformity of color level (LAB value)	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Each value (max-min)/(2*avg)<10%
Ion contamination	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	< 4.5 ug NaCl/sq.in

8.3.1.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Hot Oil
- Reflow
- Insulation resistance (trace to trace)
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.1.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.2.Hot press laminator

8.3.2.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Dielectric thickness uniformity	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Thickness (max-min)/(2*avg)<10%
Void in dielectric	Working Panel	Visual inspection 1 working panels/lot by etching out the Cu foil	No void
Board thickness	Working Panel	Check 100% panels for all the lots	Cpk>1.33
Temperature profile on actual panel	Test Coupon	Check top, middle and bottom actual panels temperature profile in each open	Temperature (max-min)/(2*avg)<10%
Butter coat thickness	Working Panel	Check top/bottom side and corner/center side butter coat, 3 panels/lot	Thickness >3um

8.3.2.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Hot Oil
- Reflow
- Insulation resistance (via to via)
- Insulation resistance (trace to trace)
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.3.Sub material cut for FPC, LCP-FPC

8.3.3.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Cosmetic	Working Panel	100% cosmetic checking for all the lots	Within p chart control limit
Punch hole position accuracy	Working Panel	Check 10 sheets accuracy by lot	Cpk>1.33
Sheet dimension	Working Panel	Check 10 sheets length/width by lot	Cpk>1.33

8.3.3.2.CQRA

Not required.

8.3.3.3.QMS

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8.3.4.FPC stacking / multilayer lay-up

8.3.4.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Ultra sonic strength (if ultrasonic bonding is used in bare FPC vendor process)	NA	Use ultra sonic pressure meter to check at least 9 points in ultrasonic cleaning section	Within SPC control limit
Layer to layer registration	Working Panel	Check 10 sheets accuracy by lot	Spec: < minimum annular ring size, Cpk>1.33

8.3.4.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Reflow
- Heat soak
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.4.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.5.Laser half-cut, punch half-cut for decap for FPC

8.3.5.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Contamination	Working panel	Check 10 panels with visual inspection	No contamination
Burr	Working panel	Check 10 panels with Visual inspection and OMM	Not allowable
Size/Position accuracy	Working Panel	Check 10 panels accuracy by lot	No significant difference statistically
Cut depth	Working Panel	Cross section on all laser cut areas on unit (can be multiple locations depending on design) 5 pcs per panel (4 corners and center), 10 panel per lot	Flex vendor internal spec Cpk > 1.33
Electrical test	Working Panel	Check 10 panels by lot	Within p-chart limits for all failure modes including spark

8.3.5.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Reflow
- Heat soak
- Insulation resistance (layer to layer)
- Insulation resistance (trace to trace)
- bHast
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.5.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.6.PI removal for decap for FPC

8.3.6.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Contamination	Working panel	Check 10 panels with visual inspection	No contamination
PI and BS Burr	Working Panel	Check 10 panels with Visual inspection and OMM Cross section on all laser cut areas on unit (can be multiple depending on design), 5 pcs per panel (4 corners and center), 10 panel per lot	Not allowable
Size/Position accuracy	Working Panel	Check 10 panels accuracy by lot	No significant difference statistically
Yield	Working Panel	Check 10 panels per lot	Yield = 100% No PI residue, PI tearing, voids, or other cosmetic failures Squeeze out: follow QCP No rework or repair allowed
Voids under PI or BS	Working Panel	Cross section on all layer transition areas on unit (can be multiple locations depending on design) 5 pcs per panel (4 corners and center), 10 panel per lot	No voids

8.3.6.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Reflow
- Heat soak
- Insulation resistance (layer to layer)
- Insulation resistance (trace to trace)
- bHast
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.6.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.7.Cover layer attach+pressing

8.3.7.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Resin flow	Working Panel	Check 10 sheets by OMM	Max 100um
Adhesion	Test coupon	Check 32 coupons by IPC TM-650 2.4.9	Material specification
Position accuracy	Working Panel	Check 10 sheets accuracy by lot	Spec: <minimum clearance, Cpk>1.33
Temperature profile on actual panel	Test Coupon	Check top, middle and bottom actual panels temperature profile in each open	Temperature (max-min)/(2*avg)<10%
Void in adhesive, or between adhesive and PI/ trace	Working Panel	Minimum 2 cross sections (0° and 90°) at 5 locations (4 corners + 1 center) in 1 working panel/lot (narrowest spaces on outer layer, and decap opening edge)	No void
Butter coat thickness	Working Panel	Check top/bottom side and corner/center side butter coat, 3 panels/lot	Thickness >3um

8.3.7.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Reflow
- Heat soak
- Insulation resistance (trace to trace)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.7.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.8.Cavity cut, outline cut, singulation

8.3.8.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Contamination	Working panel	Check 10 sheets with visual inspection	No contamination
Burr	Working panel	Check 10 sheets with Visual inspection and OMM	Not allowable
Size/Position accuracy	Working Panel	Check 10 sheets accuracy by lot	No significant difference statistically

8.3.8.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Reflow
- Heat soak
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.8.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.3.9.Stiffener/EMI shielding attach+baking

8.3.9.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Resin flow	Product	Check 32pcs by Cross section	Meet MCO requirement and no squeeze out of stiffener outline
Adhesion, peel strength	Working panel	Check 10 sheets by lot. **EMI shielding**: use 3M tape testing. **Stiffeners**: peel the stiffener itself	No peeling **For FPC**: refer to QCP 080-03022
Size/Position accuracy	Working Panel	Check 10 sheets accuracy by lot	No significant difference statistically
Warpage (not applicable for EMI shielding)	Working Panel	Check 10 sheets warpage by lot, or actual parts after singulation	Cpk>1.33
Grounding resistance (applicable for SUS stiffener only)	Working Panel	Check 10 sheets/lot resistance from before reflow and after 4X reflow	No significant difference statistically

8.3.9.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Reflow
- Heat soak
- Insulation resistance (layer to layer)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.3.9.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.4.Soldermask (LPI) formation

8.4.1.Pre-treatment for Solder mask coating

8.4.1.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Roughness uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Roughness (max-min)/(2*avg)<10%

8.4.1.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Insulation resistance (trace to trace)
- Flux resistance
- High speed ball shear test
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.4.1.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.4.2.Soldermask coating

8.4.2.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Solder mask thickness uniformity (after post cure)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Thickness (max-min)/(2*avg)<10%
Solder mask height difference between SMT pads surface and SM surface (after post cure, for both SMD and NSMD pads)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Solder mask thickness over the trace corner (after post cure)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33

8.4.2.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Insulation resistance (trace to trace)
- Flux resistance
- High speed ball shear test
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.4.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.4.3.Soldermask Pre-curing

8.4.3.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Solder mask opening size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33

8.4.3.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency) for **FPC**.

- Insulation resistance (trace to trace)
- Flux resistance
- High speed ball shear test
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**

8.4.3.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.4.4.Soldermask Exposure

8.4.4.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Solder mask opening size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Coverage (distance between SM edge and nearest covered trace)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk > 1.33 for coverage > 5μm
Alignment accuracy	Test Panel	Use test panel to check alignment accuracy by lot	Spec: <minimum clearance size, Cpk>1.33

8.4.4.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Insulation resistance (trace to trace)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- High speed ball shear test
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness

8.4.4.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.4.5.Soldermask Developing

8.4.5.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Solder mask opening size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Coverage (distance between SM edge and nearest covered trace)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk > 1.33 for coverage > 5μm
Cosmetic yield	Working Panel	100% visual inspection	Within p chart control limit

8.4.5.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Insulation resistance (trace to trace)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- High speed ball shear test
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness

8.4.5.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.4.6.Soldermask Post thermal curing, UV curing

8.4.6.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Solder mask thickness uniformity	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Thickness (max-min)/(2*avg)<10%
Solder mask height difference between SMT pads surface and SM surface (after post cure, for both SMD and NSMD pads)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Solder mask thickness over the trace corner	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Tape peeling test	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	No peeling

8.4.6.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Insulation resistance (trace to trace)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- High speed ball shear test
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness

8.4.6.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.4.7.Soldermask Plasma cleaning

8.4.7.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Solder mask and Cu pad surface energy	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Within control limit
Tape peeling test	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	No peeling

8.4.7.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Insulation resistance (trace to trace)
- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- High speed ball shear test
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness

8.4.7.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.5.Surface finish formation

8.5.1.Pre-treatment for ENIG plating

8.5.1.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Cosmetic yield	Working Panel	100% visual inspection for all the lots	Within p chart control limit
NSMD Pad size	Working Panel	Check top/bottom side and operation/center/pump side by lot	Cpk>1.33

8.5.1.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- High speed ball shear test
- Soldermask / plating adhesion (No sign of peeling)

8.5.1.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.5.2.ENIG plating

8.5.2.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and corner/center side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Cosmetic yield	Working Panel	100% visual inspection for all the lots	Within p chart control limit
NSMD Pad size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33
Ni/Au thickness	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33
Tape peeling test	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	No peeling
SEM for Ni grain structure (Au removed)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	No Ni grain boundary corrosion
FIB for AU/Ni/Cu interface	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	No void between each metal interface, Spike
Roughness of ACF pad (FPC)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Within OK2Ship spec

8.5.2.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness
- Solder ability test
- High speed ball shear test

8.5.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.5.3.Soft direct Au plating

8.5.3.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and corner/center side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Cosmetic yield	Working Panel	100% visual inspection for all the lots	Within p chart control limit
NSMD Pad size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Au thickness	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Tape peeling test	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	No peeling
Roughness	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Within OK2Ship spec

8.5.3.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness
- Solder ability test
- High speed ball shear test

8.5.3.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.5.4.Hard Au plating

8.5.4.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and corner/center side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Cosmetic yield	Working Panel	100% visual inspection for all the lots	Within p chart control limit
NSMD Pad size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Au thickness	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Hardness	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Within OK2Ship spec

8.5.4.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness
- Solder ability test
- High speed ball shear test

8.5.4.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.5.5.ENEPIG

8.5.5.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and corner/center side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Cosmetic yield	Working Panel	100% visual inspection for all the lots	Within p chart control limit
NSMD Pad size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Ni/Pd/Au thickness	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
Tape peeling test	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	No peeling
FIB for Au/Pd/Ni/Cu interface	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	No void between each metal interface

8.5.5.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Soldermask hardness
- Solder ability test
- High speed ball shear test

8.5.5.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.5.6.OSP

8.5.6.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
Cosmetic yield	Working Panel	100% visual inspection for all the lots	Within p chart control limit
NSMD Pad size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/ lot	Cpk>1.33
OSP thickness	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Cpk>1.33

8.5.6.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Thermal stability (Solder shock) for **rigid and rigid-flex PCB**, thermal stress for **FPC**
- Flux resistance
- Chemical resistance
- Soldermask / plating adhesion (No sign of peeling)
- Ionic contamination
- Solder ability test
- High speed ball shear test

8.5.6.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.6.Inspection and others

8.6.1.AFVI, VRS, artificial intelligence

8.6.1.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Gage R&R	Shipping Strip	32 strips/lot	Attribute GR&R: Kappa>0.9
Correlation between 100% 40X micro scope checking and AVI	Shipping Strip	32 strips/lot	No escaping
FVI false alarm rate	Shipping Strip	32 strips/lot	False error quantity / true failure quantity < 150%
AI filter rate	Working panel	Review overall filter rate	Filter rate > 50%
AI escape rate	Working panel	Compare defect quantity captured by 1) VRS after AVI using AI and 2) VRS after AVI not using AI (including ODB++ mismatch)	• No escape from AI. Check each defect type. • Labeled picture > 1K per defect type. Pass pictures : total defect picture ~ 50 : 50. Overall inspected quantity > 100K.
AI score plot	Working panel	Review AI score distribution of binary classification & multiple classification	• Min 1K pictures per defect type in trained database & total 100k pictures inspected for qualification are needed. • AI threshold > highest NG score + 1 sigma in binary classification. • AI threshold > highest of other classification score + 1 sigma in multiple classification.
Kappa	Shipping Strip	Kappa, κ= (Po-Pe)/(1-Pe) 5 pcs serious defect + 5pcs slight defect per defect type + 20pcs good pcs.	κ > 0.9 No defect escape

Definitions:

- Filter rate = quantity judged by AI correctly / suspected NG quantity judged by AOI or AVI
- Escape rate = defect quantity escaped from AI / overall AI inspection quantity
- Po is the relative observed agreement among raters, and Pe is the hypothetical probability of chance agreement

8.6.1.2.CQRA

Not required.

8.6.1.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC, defect type categories of artificial intelligent, AI score plot

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8.6.2.ET (O/S)

8.6.2.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Gage R&R	Shipping Strip	32 strips/lot	Attribute GR&R: Kappa>0.9 ANOVA GRR: <30%
Resistance correlation with golden machine	Shipping Strip	32 strips/lot	Two sample t-test: R square>90% Matched pairs: no significant difference statistically (the confidence interval must contain zero, and t-tests p-value must be >5%)
Cosmetic	Shipping Strip	100% visual inspection	No pin mark
Match e-map OK/NG records	Shipping Strip	32 strips/lot	Compare log from tester to e-map at the end of the process. Confirm that e-map has registered all e-test failures as NG.

8.6.2.2.CQRA

Not required.

8.6.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.6.3.DCR, Impedance (TDR)

8.6.3.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Impedance correlation with golden machine	Impedance coupon	32 coupons/lot	Two sample t-test: R square>90% Matched pairs: no significant difference statistically (the confidence interval must contain zero, and t-tests p-value must be >5%)
Cosmetic	Shipping Strip	100% visual inspection	No pin mark

8.6.3.2.CQRA

Not required.

8.6.3.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.6.4.ICT for FPC, LCP-FPC

8.6.4.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Gage R&R	Shipping Strip	32 strips/lot	Attribute GR&R: Kappa>0.9 ANOVA GRR: <30%
Correlation with golden machine	Shipping Strip	32 strips/lot	Two sample t-test: R square>90% Matched pairs: no significant difference statistically (the confidence interval must contain zero, and t-tests p-value must be >5%)
Cosmetic	Shipping Strip	100% visual inspection	No pin mark

8.6.4.2.CQRA

Not required.

8.6.4.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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8.6.5.2DID, Mark printing for FPC, LCP-FPC

8.6.5.1.Product Quality and Capability confirmation

Item	Specimen	Test and Sampling Method	Judgment criteria
Readability	Shipping Strip	100% reader checking	grade B or above
Tape peeling test	Shipping Strip	32 strips/lot	No peeling
Cosmetic	Shipping Strip	100% visual inspection	No contamination caused by marking
Registration	Shipping Strip	Check all the 2DID, printed marks on 3 working panels / lot by OMM	Cpk > 1.33 with internal spec
Link to traceability system	Shipping Strip	Check all the 2DID, printed marks on 3 working panels / lot by OMM	All 2DID, printed marks refer to traceability data

8.6.5.2.CQRA

Test items as below. For test methods and sampling plan, refer to 099-01674 for **rigid and rigid-flex PCB**, and CQRA 080-02999 and QCP 080-03022 (NPI level frequency), and specifications 080-2265 for **FPC**.

- Solder-ability test
- Ionic contamination
- High speed ball shear test

8.6.5.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9. Evaluation and qualification items for materials

9.1.Materials for FPC

9.1.1.FCCL

9.1.1.1.Product Quality and Capability confirmation

Process to qualify as below. Refer to appropriate process section above for UPD items to follow for qualification of new FCCL material.

- Laser via drilling
- Plasma dry desmear / liquid desmear
- Overhang Cu flash etching/Cu foil etching
- Electroless Cu plating
- Shadow/Black hole
- Flash Cu plating
- Pre-treatment for Dry film lamination
- Dry film exposure, Direct imager for “Cu trace”
- Electrolytic Cu pattern/via-fill plating
- Via Paste Filling
- Subtractive Development-Etching-Stripping
- Cavity cut, outline cut, singulation

9.1.1.2.CQRA

Test items as below. Refer to QCP (080-03022) for sampling plan requirements (NPI level frequency), and CQRA (080-02999) document for test methods. Continues on next page.

- Reflow
- Solderability Test
- Thermal Stress
- Hot bar loop test
- Hot Oil Test
- Insulation Resistance (Via to Via)
- Insulation Resistance (Layer to Layer)
- Ionic Contamination
- Dielectric Withstanding

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- Moisture absorption
- Thermal cycling
- Heat Soak and Recovery
- Thermal Shock
- Flex bending test
- Thermal Cycling and Flex Bending
- Heat Soak and Flex Bending
- FIB/IM test
- bHAST
- Survival Test-Reflow
- Survival Test-Hot Oil
- High Speed Ball Shear

9.1.1.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9.1.2.Coverlay

9.1.2.1.Product Quality and Capability confirmation

Process to qualify as below. Refer to appropriate process section above for UPD items.

- Coverlay attach + pressing

9.1.2.2.CQRA

Test items as below. Refer to QCP (080-03022) for sampling plan requirements (NPI level frequency), and CQRA (080-02999) document for test methods.

- Reflow
- Thermal Stress
- Hot bar loop test
- Ionic Contamination
- Moisture absorption
- Thermal cycling
- Heat Soak and Recovery
- Thermal Shock
- Flex bending test
- Thermal Cycling and Flex Bending
- Heat Soak and Flex Bending
- bHAST
- Survival Test-Reflow

9.1.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9.1.3.Bonding sheet

9.1.3.1.Product Quality and Capability confirmation

Process to qualify as below. Refer to appropriate process section above for UPD items.

- Laser via Drilling
- Plasma dry desmear/Liquid Desmear
- Overhang Cu flash etching/Cu foil etching
- Electroless Cu plating
- Shadow/Black hole
- Flash Cu plating
- Dry film exposure, Direct imager for “Cu trace”
- Electrolytic Cu pattern/via-fill plating
- Via Paste Filling
- Brown oxide for dielectric lamination
- Hot press lamination
- Laser half-cut, punch half-cut for decap
- PI removal for decap

9.1.3.2.CQRA

Test items as below. Refer to QCP (080-03022) for sampling plan requirements (NPI level frequency), and CQRA (080-02999) document for test methods. Continues on next page.

- Reflow
- Thermal Stress
- Hot bar loop test
- Hot Oil Test
- Insulation Resistance (Via to Via)
- Insulation Resistance (Layer to Layer)
- Ionic Contamination
- Dielectric Withstanding
- Moisture absorption
- Thermal cycling
- Heat Soak and Recovery
- Thermal Shock
- FIB/IM est

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- bHAST
- Survival Test-Reflow
- Survival Test-Hot Oil

9.1.3.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9.1.4.Solder mask

9.1.4.1.Product Quality and Capability confirmation

Process to qualify as below. Refer to appropriate process section above for UPD items.

- Solder mask coating
- Solder mask Pre-curing
- Solder mask Exposure
- Solder mask Developing
- Solder mask Post thermal curing, UV curing
- Solder mask Plasma cleaning

Additional check items as below.

Process	Product Quality and Capability Confirmation	Specimen	Test and Sampling Method	Judgment criteria
Leaching	Au Grain surface SEM, Au removed Ni grain surface SEMM	Test coupon	1pcs/lot	No black nickel, corrosion spike size should be <1μm or 20% of Ni thickness
Hardness	Solder Mask Hardness	-	Based on 099-01674	Based on 080-2265

9.1.4.2.CQRA

Test items as below. Refer to QCP (080-03022) for sampling plan requirements (NPI level frequency), and CQRA (080-02999) document for test methods. Continues on next page.

- Reflow
- Solderability Test
- Thermal Stress
- Hot bar loop test
- Ionic Contamination
- Solder Mask/Plating Adhesion
- Chemical Resistance
- Flux Resistance
- Thermal cycling
- Heat Soak and Recovery

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- Thermal Shock
- bHAST
- Survival Test-Reflow
- Survival Test-Hot Oil
- High Speed Ball Shear

9.1.4.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9.1.5.EMI shield

9.1.5.1.Product Quality and Capability confirmation

Process to qualify as below. Refer to appropriate process section above for UPD items.

- EMI shielding attach+baking

9.1.5.2.CQRA

Test items as below. Refer to QCP (080-03022) for sampling plan requirements (NPI level frequency), and CQRA (080-02999) document for test methods.

- Reflow
- Thermal Stress
- Hot bar loop test
- Ionic Contamination
- Moisture absorption
- Thermal cycling
- Heat Soak and Recovery
- Thermal Shock
- Flex bending test
- Thermal Cycling and Flex Bending
- Heat Soak and Flex Bending
- bHAST
- Survival Test-Reflow

9.1.5.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9.2.Materials for rigid PCB

9.2.1.Pre-preg

9.2.1.1.Product Quality and Capability confirmation

Process	Product Quality and Capability Confirmation	Specimen	Test and Sampling Method	Judgment criteria
Laser Drilling	Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%
	Wedge	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Wedge<10um
Plasma dry desmear /Liquid Desmear	Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/ (2*avg)<10%
	Surface discoloration	Working Panel	Check 3 working panels/lot	No scratch, discoloration, water stain
	Uniformity of color level (LAB value)	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Each value (max-min)/ (2*avg)<10%
	Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Overhang<10um
Electroless Cu plating	Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/ (2*avg)<10%
	FIB to check E'less Cu thickness on via wall	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	Follow 062-0073, >0.2um
	E'less Cu plating thickness on surface (Weight gain)	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Plating thickness or weight gain (max-min)/(2*avg)<10% and within SPC control limit
Shadow	Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/ (2*avg)<10%
	SEM/FIB after Cu plating to check the bottom of via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 8 laser via/lot at least.	No voids, carbon residue

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Process	Product Quality and Capability Confirmation	Specimen	Test and Sampling Method	Judgment criteria
/ Black hole	Resistance checked by via chain coupon	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Resistance (max-min)/ (2*avg)<10% and within SPC control limit
	Via Hole AOI/AOM	Working Panel	100% AOI and minimum 10% AOM	
AOI/VRS	Defect escaping	Working Panel	Check the grey level setting and actual 10 panels/lot escaping rate	Zero escaping
Hot press laminator	Dielectric thickness uniformity	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Thickness (max-min)/ (2*avg)<10%
	Void in dielectric	Working Panel	Visual inspection 1 working panels/lot by etching out the Cu foil	No void
	Board thickness	Working Panel	Check 100% panels for all the lots	Cpk>1.33
	Temperature profile on actual panel	Test Coupon	Check top, middle and bottom actual panels temperature profile in each open	Temperature (max-min)/ (2*avg)<10%
	Butter coat thickness	Working Panel	Check top/bottom side and corner/center side butter coat, 3 panels/lot	Thickness >3um

9.2.1.2.CQRA

Test items as below. Refer to Quality Assurance and Control of Rigid and Rigid Flex PCB (099-01674) for test methods.

- Hot Oil
- x20 Reflow
- Insulation resistance (Via to Via)
- Insulation resistance (Trace to Trace)
- Insulation resistance (Layer to Layer)
- Thermal stability (Solder shock)

9.2.1.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9.2.2.Solder mask ink / Dry Film

9.2.2.1.Product Quality and Capability confirmation

Process	Product Quality and Capability Confirmation	Specimen	Test and Sampling Method	Judgment criteria
Solder mask Pre-curing	Solder mask thickness unformility	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33
Solder mask Exposure	Solder mask opening size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33
Solder mask Developing	Undercut	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	<15um, Cpk>1.33
	Solder mask opening size	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33
	Undercut	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	<15um, Cpk>1.33
	Cosmetic yield	Working Panel	100% visual inspection	Within p chart control limit
Solder mask Post thermal curing, UV curing	Solder mask thickness uniformity	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Thickness (max-min)/ (2*avg)<10%
	Solder mask height difference	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33
	Solder mask thickness over the trace corner	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Cpk>1.33
Solder mask Plasma cleaning	Solder mask and Cu pad surface energy	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Within control limit

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Process	Product Quality and Capability Confirmation	Specimen	Test and Sampling Method	Judgment criteria
ENIG	ENIG leaching test and check Au Grain surface SEM, Au removed Ni grain surface SEMM	Test coupon	1pcs/lot	No black nickel, corrosion spike size should be <1μm or 20% of Ni thickness
	Tape peeling test (After ENIG)	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	No peeling
	Solder Mask Hardness (After ENIG)	Working Panel	Based on 099-01674	> 6H
	Water contact angle	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Statistically equal with POR or meet Apple defined spec
	L*AB	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Statistically equal with POR or meet Apple defined spec
	Gloss unit	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Statistically equal with POR or meet Apple defined spec

9.2.2.2.CQRA

Test items as below. Refer to Quality Assurance and Control of Rigid and Rigid Flex PCB (099-01674) for test methods.

- Insulation resistance(Trace to Trace)
- Thermal stability (Solder shock)
- Flux Resistance
- Chemical Resistance
- Solder Mask/Plating Adhesion
- Ionic Contamination
- Solder Mask Hardness

9.2.2.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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9.2.3.Copper foil

9.2.3.1.Product Quality and Capability confirmation

Process	Product Quality and Capability Confirmation	Specimen	Test and Sampling Method	Judgment criteria
IQC	Morphology on the side where will be attached to dielectric	Sheet	Check corner/center of sheet by SEM, 3 sheets	Gross Reality Check, comparison to POR with no observable differences that would fail engineering judgement
	Ra/Rz on the side where will be attached to dielectric	Sheet	Check corner/center of sheet, 10 sheets	If intended to be different for lower or higher roughness, then statistical difference to POR is required, otherwise statistical equivalence to POR if being qualified as a lower cost or alternative.
Hot press laminator	Cu foil thickness uniformity	Working Panel	Check top/bottom side and corner/center side for uniformity, 3 panels/lot	Thickness (max-min)/(2*avg)<10%
	Cu foil peel strength	Test Coupon	Check top/bottom side, 3 panels/lot	Follow 062-0073
	mSAP Cu foil carrier miss-peeling rate	Working Panel	Check 100% panels for all the lots	Defect rate <0.2%
Patterning etching process (subtractive and mSAP)	Etching amount and uniformity on Cu foil	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
	Trace width, pad size	Working Panel	Check top/bottom side and operation/center/pump side by lot	Cpk>1.33
	AOI yield	Working Panel	100% AOI for all the lots	Within AOI p chart control limit
	Etching amount and uniformity	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Etching amount (max-min)/(2*avg)<10%
	Surface discoloration	Working Panel	Check 3 working panels/lot	No scratch, discoloration, water stain

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Process	Product Quality and Capability Confirmation	Specimen	Test and Sampling Method	Judgment criteria
Laser pre-treatment	Uniformity of color level (LAB value)	Test Coupon	Check top/bottom side and operation/center/pump side for uniformity by lot	Each value (max-min)/ (2*avg)<10%
	Overhang (or etching back in FPC) amount on laser via	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Overhang<10um
	Top and bottom via size, open ratio (bottom/top)	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Via size Cpk>1.33, open ratio>80%
	Wedge	Working Panel	Cross section top/bottom side, corner/center position and highest/lowest dielectric thickness(highest/lowest Cu% area). 32 laser via/lot at least.	Wedge<10um

9.2.3.2.CQRA

Test items as below. Refer to Quality Assurance and Control of Rigid and Rigid Flex PCB (099-01674) for test methods.

- Hot Oil
- x20 Reflow
- Insulation resistance (Via to Via)
- Insulation resistance (Trace to Trace)
- Insulation resistance (Layer to Layer)
- Thermal stability (Solder shock)

9.2.3.3.QMS

SOP, Product quality check sheet, Equipment check sheet, First article system, Shop floor system, Process parameter settings, Chemical management, SPC

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Appendix A: Revision History

Revision	ECO	Released date	Released by	Description
A	0016627610	Jan 29, 2019	Ian Tsai	Initial release
B	0019648539	Jun 26, 2019	Ian Tsai	P.4/5 update purpose statement, add “qualification scenario”, “risk assessment”, “new method and tool exception”, “equipments/chemicals list” Add P.8 “existing equipment qualification procedure” Add P.9 “equipments/chemicals list template” P.15/16 add SEM/EDX from laser via top view after desmear Add P.27 for via paste filling
C	0021310108	Nov 25, 2019	Ian Tsai	1. Add material qualification
D	0022615830	Feb 25, 2020	Benoit Huneau	Clarified qualification scenario throughout the builds
E	0023391845	Apr 24, 2020	Ian Tsai Benoit Huneau	Added Cu foil material qualification for Rigid PCB Re-organized conditions for qualification scenario and qualification procedure. Reduced qualification steps requirements to be UPD step 1 only for some scenario
F	0023811137	May 14, 2020	Benoit Huneau	Updated wording, added scenario and updated criteria and requirement in the table of scenario on page 7 Removed unnecessary comments on page 9 Corrected the “9 lots” typo on individual process qualification criteria
G	0028559508	Apr 13, 2021	Benoit Huneau Ian Tsai	P.7: Added scenario 9 with different panel size at P2, and updated few other items P.9&10: Aligned milestone numbering P.10: Added the quantity of lots for final qualification P.35: Added different material test requirements in AOI/VRS qualification P.44: Updated criteria P. 61~64: Clarified criteria for GRR and correlation for tester qualification P.76: Added AOI VRS grey level checking and zero defect escaping requirement

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Revision	ECO	Released date	Released by	Description
H	0030303789	Jul 12, 2021	Benoit Huneau	• Changed formatting • **6.1. qualification unit**: clarified requirement to use one type of machine • **6.5. equipments list**: updated requirements • **7.1. qualification scenario for equipments and methods**: clarified requirements • **7.3. workflow for new materials**: simplified flow • **8. Evaluation and Qualification Items for process**: Replaced CQRA test conditions for rigid and rigid-flex details to reference to 099-01674 • **8.1.3. laser via drilling**: added SEM/EDX check of via bottom • **8.3.7. laser half-cut, punch half-cut for decap**: added • **8.3.8. PI removal for decap**: added • **8.6. inspections & others section**: added validation with e-map comparison • **8.6.5. 2DID, Mark printing**: updated UPD and CQRA requirements • **9.1. Materials for FPC**: removed overlapping requirements.
J	0034158574	Mar 14, 2022	Ian Tsai Benson YL Lin Benoit Huneau	• **6.1. Qualification Unit**: clarified vendor's responsibilities • **7.1. Procedure for equipments and methods**: Clarified need to highlight in case of schedule risks. • **7.2. Workflow for equipments and methods**: Added PRQ steps • **7.3. Workflow for material**: split scenarios of new material qualification and material change qualification • **8.3. Evaluation and Qualification for process**: clarified thermal stability conditions for FPC • **8.3.9 Cleaning after cavity cut/outline cut/singulation**: added guidelines • **8.4. Solder mask**: added flux resistance and high speed ball shear for CQRA testing • **8.6.1 AFVI, VRS, artificial intelligent**: added items related to AI qualification • **9.1.5. new EMI shield material**: added

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